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Chapter 2

Initial Feasibility Study

2.1 Introduction

This chapter presents the initial feasibility study of the PoC-RUET directory application. The chapter examines the practicality of the app as a digital campus directory and communication support system for students, faculty members, officers, and staff of Rajshahi University of Engineering & Technology (RUET). Identify early-stage issues that influence the usefulness, accessibility, trustworthiness, and long-term sustainability of the system in a real university environment.

The purpose of this chapter is to evaluate whether the current concept and feature set of the app are sufficient to support day-to-day institutional communication. By analyzing the existing limitations of the directory from both technical and operational perspectives, this feasibility study highlights the most important challenges that may reduce user adoption, weaken confidence in the system, and limit the app's ability to function as a reliable campus-wide service.

2.2 Initial Feasibility Analysis

The initial feasibility analysis is used to assess whether the PoC-RUET app can realistically operate as a reliable, scalable, and user-friendly university contact directory. This analysis considers the present architecture, expected user behavior, maintenance demands, and institutional requirements to determine whether the application is practical for broader implementation and future improvement.

A preliminary review of the PoC-RUET directory application indicates that the system has significant promise, but its success depends on how effectively a set of core limitations is addressed. The feasibility assessment can be understood through the following dimensions:

1. **Technical Feasibility:** The app must maintain accurate, searchable, and consistently available directory information while performing well under real campus conditions, including weak connectivity, frequent data changes, and growing data volume.
2. **Economic Feasibility:** The system should reduce repetitive manual work, moderation overhead, and administrative dependency so that the cost of maintaining and scaling the platform remains reasonable.
3. **Operational Feasibility:** The platform must support the needs of a diverse user base, including faculty members, students, officers, and lower-level support staff, without introducing barriers to access or use.
4. **Legal and Ethical Feasibility:** Profile information, communication data and uploaded photos must be managed responsibly, taking into account privacy, verification, and institutional policy.
5. **Schedule Feasibility:** Enhancements should be implementable in manageable phases so that the most urgent weaknesses can be addressed first without delaying the overall progress of the application.

2.2.1 Problem 01

Statement of the problem

Data accuracy and stale information within the directory represent a major problem in the PoC-RUET app. This issue occurs when staff details change at the institutional level, but the app database remains static or outdated, ultimately affecting the reliability and credibility of the entire system.

Summary of findings

Users may encounter incorrect phone numbers, outdated designations, inaccurate office locations, or incomplete staff records because administrative changes are not reflected in the app quickly enough. As a result, the directory may not provide reliable contact information at the time it is needed.

Details of findings

In a university environment, personnel information changes frequently due to promotions, transfers between departments, role adjustments, retirements, and new appointments. When the PoC-RUET

app is dependent on manual data entry or delayed batch updates, the system can quickly fall behind the current institutional record. Even a small delay between an official administrative change and its appearance in the app can create confusion for users trying to contact a staff member urgently.

This problem weakens the practical value of the platform because the usefulness of a directory depends primarily on the correctness of its information. If users repeatedly encounter invalid numbers or outdated office assignments, they begin to question the dependability of the application as a whole. Over time, this can force students, staff, and visitors back to manual inquiries, departmental phone calls, or informal social channels, reducing the effectiveness of the app as a centralized digital directory.

Recommendations

Automated synchronization should be implemented between the PoC-RUET database and the central university HR or payroll systems so that official changes can be reflected in the app with minimal delay. In addition, a user-facing “Suggest an Edit” or “Update My Profile” feature should be introduced so that staff members can flag inaccuracies or request profile corrections directly. Together, these two measures would reduce update latency, improve data integrity, and help maintain user trust in the platform.

2.2.2 Problem 02

Statement of the problem

Rigid search functionality and filtering limitations create another important problem within the app. This issue appears in the main search interface, where users often struggle to locate contacts unless they already know the exact name, spelling, or formal designation of the person they are searching for.

Summary of findings

Users often fail to find the intended contact when they misspell a name, search by a partial title, or rely on a responsibility-based query rather than an exact identity match. The current behavior suggests that the search process is too dependent on strict string matching and lacks flexible discovery support.

Details of findings

People with different levels of familiarity with campus personnel. New students, guardians, visitors, and even members of other departments may know only partial information about a person. For instance, a user may search for “CSE Head,” “Dean Engineering,” or a name with a minor spelling error, expecting the system to infer the intended result. If the app returns no meaningful matches in such situations, the search experience becomes frustrating and inefficient.

This limitation reduces the accessibility of the directory because the burden of precision is placed entirely on the user. Instead of helping users find the correct person through approximations, role-based search, or filtered exploration, the app behaves like a narrow database lookup tool. As a result, users who do not know the exact spelling or formal designation may conclude that the desired person is absent from the system, even when the record is actually present.

Recommendations

The app should incorporate fuzzy search logic so that minor spelling errors, phonetic variations, and partial matches can still return relevant results. Advanced filtering options should also be added to allow users to search by research interests, laboratory affiliation, administrative responsibilities, academic departments, or role categories. These improvements would make the search process more inclusive, more intuitive, and better aligned with real user behavior.

2.2.3 Problem 03

Statement of the problem

An architectural limitation causing a lack of offline functionality is another significant concern. Currently, the app depends heavily on a continuous online connection and does not yet provide built-in offline caching, making it unavailable in areas with poor reception.

Summary of findings

The app may fail to load contact data or search results in known dead zones of the campus, such as basement areas, deep laboratory spaces, or locations with weak Wi-Fi or unstable mobile signals. This means users must depend on an active and stable connection at all times.

Details of findings

Campus connectivity is not always consistent, especially in enclosed buildings, remote corners of departments, or during periods of network congestion. In such conditions, a fully cloud-dependent directory becomes unreliable at the exact moments when quick access to information may be most important. A student may need an office room urgently, a visitor may need help finding a staff member, or a user may need immediate access to an emergency contact number. If the app cannot respond without internet access, its real-world effectiveness becomes severely limited.

This issue affects not only convenience but also system resilience. A directory app that becomes unusable during connectivity interruptions cannot fully support a campus environment where users move across buildings and rely on the app in unpredictable conditions. The absence of offline capability therefore reduces both accessibility and service continuity.

Recommendations

Local data caching should be prioritized in future development. The app should download a compressed and encrypted version of essential directory data, including names, phone numbers, and office locations, onto the user's device for offline querying. Whenever the device reconnects to a stable network, the data can be synchronized silently in the background. This approach would ensure that the most critical directory features remain available even in low-connectivity environments.

2.2.4 Problem 04

Statement of the problem

An inclusion gap for lower-level support staff presents a structural problem in the system. It is currently difficult to integrate 3rd and 4th-class employees cleanly into the central digital directory when access depends on the same formal identity and authentication structure used for other groups.

Summary of findings

Lower-tier support staff may be absent from the directory or unable to access the app because the authentication and profile management process appears too rigid for their practical circumstances.

Details of findings

If the PoC-RUET app relies heavily on official `ruet.ac.bd` email-based single sign-on, it may unintentionally exclude important categories of employees such as guards, cleaners, attendants, drivers, and lab assistants. Some of these staff members may not receive institutional email addresses, while others may have limited digital literacy or may not be comfortable completing an email-led registration flow. As a result, a part of the university workforce becomes underrepresented or entirely invisible in the digital directory.

This creates both an inclusiveness problem and a practical communication gap. Support staff often play vital operational roles within departments and offices, and their absence from the system weakens the completeness of the directory. In a campus service intended to improve institutional communication, excluding one segment of the workforce reduces the value of the platform for everyone else as well.

Recommendations

An alternative authentication method, such as OTP verification through registered mobile numbers, should be introduced for specific user tiers that may not have institutional email access. In addition, authorized administrative officers should be able to create and manage read-only profiles for department support staff so that their contact information remains searchable even when they do not actively use the app themselves. Such measures would improve inclusivity and strengthen the completeness of the university directory.

2.2.5 Problem 05

Statement of the problem

Technical debt in visual data management is another notable issue in the PoC-RUET app. Significant difficulties arise when older uploaded profile photos must be managed, moderated, and verified manually over time.

Summary of findings

The existing process of manually checking user-uploaded photos to ensure that they appropriately match the employee or student creates an administrative bottleneck that becomes increasingly difficult to sustain as the system grows.

Details of findings

As the directory expands, manual review of legacy image data becomes time consuming and inefficient. Older profile images may be low in resolution, poorly cropped, outdated, visually inconsistent, or unsuitable for use in an official university directory. Without an efficient validation mechanism, the system accumulates visual inconsistency and moderation backlog, which increases the burden on the administrators responsible for maintaining the app.

This issue affects more than appearance alone. Profile photos contribute to recognition, trust, and professionalism, especially in a university setting where users may rely on images to identify faculty members, officers, or staff members correctly. If image quality varies widely or if verification is delayed, the app may appear less credible and less professionally maintained. Over time, this can reduce confidence in the quality of the platform as a whole.

Recommendations

Automated image validation should be introduced during the upload process. For example, the system may check whether a face is present, whether the image meets a minimum resolution threshold, and whether the photo follows basic profile standards. To reduce the accumulated technical debt, the university may also launch a one-time photo update campaign requiring users to re-upload compliant images. Final approval can then be delegated to departmental authorities rather than centralized entirely at the ICT Cell, thereby reducing bottlenecks and improving scalability.

2.2.6 Problem 06

Statement of the problem

A disconnect between digital contact locations and physical campus navigation is also evident in the current app design. The problem occurs when users are able to identify a person digitally but still struggle to locate the corresponding office or department physically on campus.

Summary of findings

Users may see a text-based location such as “Building A, Floor 3” or a room number, but they are not given enough spatial context to understand where that place is in relation to the RUET campus layout.

Details of findings

A plain room number or building label may be understandable to regular members of a department, but it is often insufficient for freshers, visiting guests, parents, or individuals from other departments. These users may know whom they need to meet yet still waste time trying to identify the correct building, entrance, floor, or route. In such cases, the directory successfully delivers contact information but fails to support the user's full task of actually reaching the intended person.

This gap limits the app's effectiveness as an end-to-end support tool. A modern campus directory should ideally bridge digital discovery and physical navigation, especially when one of the main goals is to make university services more accessible. Without location visualization or contextual guidance, the user experience remains incomplete.

Recommendations

Contact profiles should be integrated with a digital campus map. When a user taps on an office location, the app should open an interactive map that shows the exact building position and provides simple walking guidance where appropriate. Even a basic map-based visual cue would make the app much more helpful for new users and visitors.

2.2.7 Problem 07

Statement of the problem

Communication bottlenecks arise because the app does not yet provide integrated messaging or calling features. Users can identify the correct contact, but they must still leave the app to initiate actual communication.

Summary of findings

Users may successfully find a phone number or contact profile, yet they must manually copy the number, memorize it temporarily, or switch to an external calling or messaging application to proceed.

Details of findings

This limitation introduces unnecessary friction into an otherwise useful workflow. A directory app should not only help users discover contact details but should also streamline the next step of reaching the person. When users are forced to move between apps repeatedly, the process becomes

slower and more error-prone. This is particularly inconvenient in urgent situations or when users are attempting to contact busy faculty members and officers quickly.

In addition, regular voice calls may not always be the most suitable method of communication. Some users may prefer messaging first, while others may benefit from internal communication options that are more appropriate for institutional use. If the app functions strictly as a read-only directory, then an important opportunity is lost to make communication more efficient and user-centered.

Recommendations

The app should include direct communication links where feasible. One-tap actions could allow users to place a phone call, open a messaging channel such as WhatsApp, or connect to an internal RUET IP phone or VoIP service if such infrastructure is available. These features would transform the app from a passive information repository into a more active communication support tool.

2.2.8 Problem 08

Statement of the problem

The app is currently underutilized as a proactive campus communication tool, which contributes to low daily engagement. At present, the platform mainly acts as a passive lookup service that users open only when they need a specific contact detail.

Summary of findings

The existing usage pattern suggests that users engage with the app only on an occasional basis because it offers limited value beyond contact search. As a result, the platform is not yet functioning as a regular part of campus digital life.

Details of findings

The ICT Cell has already established a potentially valuable digital channel by deploying the app to members of the university community. However, the app is not yet being used to its full strategic potential. Universities often need a fast way to communicate urgent notices, changes in schedule, weather-related closures, emergency messages, or deadline reminders. If the PoC-RUET app remains only a static directory, then it misses the opportunity to become a more central and frequently used platform.

Low engagement can also affect long-term sustainability. Users are more likely to keep an app installed, updated, and trusted when it provides recurring value. If it is used only rarely, user attention declines and the perceived importance of the platform may diminish. This makes it harder for the app to establish a strong presence within the university’s digital ecosystem.

Recommendations

The app’s existing user base should be used to support targeted push notifications for urgent and relevant announcements. These may include sudden weather closures, broad class cancellations, administrative reminders, or important deadlines linked to RUET online services. By introducing carefully managed notification features, the app could improve daily relevance while also serving as a more effective institutional communication channel.

2.3 Conclusion

In conclusion, the initial feasibility study of the PoC-RUET directory application establishes an important foundation for the later stages of system analysis, design, and development. This study demonstrates that the value of the app lies not only in providing a centralized contact directory, but also in its potential to become a dependable digital support platform for the wider RUET community. However, achieving that goal requires careful attention to the key limitations identified throughout this chapter.

The analysis has revealed several significant concerns, including stale and inaccurate directory data, limited search flexibility, the absence of offline access, exclusion risks for lower-level support staff, inefficiencies in photo verification and moderation, weak linkage between digital records and physical campus navigation, communication friction caused by the lack of integrated contact actions, and the underuse of the app as a proactive communication channel. These issues affect not only technical performance, but also user trust, inclusiveness, operational usefulness, and long-term adoption.

By addressing these challenges through targeted improvements, the PoC-RUET app can be developed into a more reliable, accessible, and user-centered system. Features such as automated data synchronization, flexible search mechanisms, offline caching, inclusive authentication methods, scalable image validation, map-assisted location guidance, integrated communication tools, and meaningful notification services would significantly strengthen the platform. Thus, this chapter has not only identified the core problems of the current concept, but has also outlined practical directions for improvement that align with both institutional needs and user expectations.